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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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DEEP LEARNING BASED REAL-TIME DRIVER DROWSINESS DETECTION AND ALERT SYSTEM

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

Course Code & Course Name

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

IN

**COMPUTER SCIENCE AND ENGINEERING
(BRANCH)**

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Deep Learning Based Real-Time Driver Drowsiness Detection and Alert System” has been carried out by D.Hasini (192425147),K.Sreelatha(192425248),V.Saiteja(192425083)and B.Jahnavi(192325043) under the supervision of DR.D Sudhagar and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, D.Hasini(192425147),K.Sreelatha(192425248),V.Saiteja(192425083)and B.Jahnavi(192325043) of the Btech.Computer Science and Engineering SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Deep Learning Based Real-Time Driver Drowsiness Detection and Alert System” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
D. Hasini (192425147)	Deep learning model design and development	Designed and trained the CNN/CNN-LSTM drowsiness classification model; tuned hyperparameters	Model-level accuracy, precision/recall and confusion-matrix evaluation	Chapter 3 (Design), Chapter 4 (Results)	25%
K. Sreelatha (192425248)	Data acquisition and preprocessing, facial landmark and eye-closure pipeline	Built the face/eye detection and Eye Aspect Ratio (EAR) feature-extraction pipeline; curated and augmented the training dataset	Preprocessing validation and feature-quality analysis	Chapter 1, Chapter 2 (Literature Review)	25%
V. Sai Teja (192425083)	Real-time video capture and hardware/alert integration	Integrated the camera module with the embedded platform and implemented the buzzer/alert triggering circuit and logic	Real-time latency, alert-response and hardware reliability testing	Chapter 4 (Implementation), Appendix B/D	25%
B. Jahnavi (192325043)	System integration, testing coordination and report documentation	Integrated software and hardware subsystems into the final prototype and coordinated overall system assembly	End-to-end system testing, test-plan design and results consolidation	Chapter 3 (Sustainability/Ethics/Risk), Chapter 5, Report compilation	25%
Total					100%

ABSTRACT

Drowsy driving remains one of the leading contributors to road traffic accidents worldwide, as a driver's ability to perceive hazards and react in time deteriorates sharply once fatigue sets in, yet existing in-vehicle safety systems rarely monitor the driver's physiological and behavioural state directly. This capstone project addresses this engineering problem by designing and implementing a Deep Learning Based Real-Time Driver Drowsiness Detection and Alert System that continuously observes a driver's facial region through a low-cost camera and issues an immediate audible-visual alert the moment signs of drowsiness are detected, so that corrective action can be taken before a lapse in attention leads to a collision.

The system need arises from the limited reach of existing solutions, which are either restricted to high-end vehicles equipped with expensive proprietary sensors or rely on indirect indicators such as steering-wheel movement and lane deviation that respond only after driving performance has already degraded. The proposed solution instead uses a vision-based deep learning pipeline: video frames are captured in real time, the driver's face and eye region are localised, and the Eye Aspect Ratio together with facial landmark features are extracted and fed to a Convolutional Neural Network (CNN) classifier trained to distinguish alert, drowsy and yawning states. A lightweight CNN architecture was selected and trained on a curated and augmented dataset of open and closed eye images to balance detection accuracy against the computational constraints of an embedded, low-cost deployment platform.

The methodology followed a standard engineering design cycle comprising requirement analysis, alternative-solution evaluation, detailed design, implementation on a Raspberry Pi with a USB camera and buzzer alert module, and systematic testing against defined acceptance criteria for detection accuracy and alert latency. The trained model achieved a classification accuracy exceeding 94% on the held-out test set, with an average end-to-end alert-response latency suitable for real-time operation, and the complete prototype was validated under varying lighting conditions and driver postures to assess robustness.

The principal conclusion of this work is that a deep learning based, camera-only drowsiness detection system can deliver dependable, low-latency alerts using inexpensive, readily available hardware, making it a practical retrofit safety solution for a wide range of vehicles rather than being limited to premium models. The project also reflects on sustainability, ethical, safety and privacy considerations associated with continuous driver-facing video monitoring, and identifies future work directions including multi-modal fusion with physiological sensors and on-device model optimisation for wider deployment.

Keywords: Driver Drowsiness Detection; Deep Learning; Convolutional Neural Network; Eye Aspect Ratio; Real-Time Monitoring; Embedded Alert System

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
EAR	Eye Aspect Ratio	—
CNN	Convolutional Neural Network	—
LSTM	Long Short-Term Memory	—
ADAS	Advanced Driver Assistance System	—
FPS	Frames Per Second	fps
GPU	Graphics Processing Unit	—
ROI	Region of Interest	—
IR	Infrared	—
ms	millisecond	ms

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Road traffic safety continues to be a major engineering and societal concern, and driver fatigue is consistently identified as a significant causal factor in a large proportion of road traffic crashes, particularly on highways and during long or night-time drives. Advanced Driver Assistance Systems (ADAS) have progressively incorporated computer-vision and deep-learning techniques to monitor both the external driving environment and, increasingly, the internal state of the driver. Advances in convolutional neural networks, facial landmark detection and low-cost embedded computing platforms (such as the Raspberry Pi and low-power GPUs) have made it technically and economically feasible to build camera-based driver-monitoring systems that were previously restricted to premium vehicles with proprietary sensor suites.

This project applies these technologies to design a Deep Learning Based Real-Time Driver Drowsiness Detection and Alert System that observes the driver's face through a standard camera, extracts drowsiness-indicative features such as eye closure duration and yawning, classifies the driver's alertness state using a trained CNN model, and triggers an immediate audio-visual alert when drowsiness is detected.

1.2 Need for the Project

Drowsy driving needs to be addressed because impaired reaction time and reduced hazard perception during microsleep episodes can directly cause loss of vehicle control, and because the affected population includes commercial long-haul drivers, shift workers, and ordinary commuters on monotonous highway stretches, all of whom are at elevated risk. Existing indirect-monitoring approaches (lane-departure warning, steering-pattern analysis) only respond after driving performance has already degraded, and vision-based commercial driver-monitoring systems remain costly and are largely confined to high-end vehicles. There is therefore a clear engineering need for an affordable, camera-only, real-time monitoring solution that directly measures physiological indicators of drowsiness rather than inferring them indirectly from vehicle dynamics.

1.3 Problem Statement

The engineering problem is to design and implement a real-time, low-cost, embedded driver-monitoring system that reliably classifies a driver's alertness state from live camera video under realistic and variable driving conditions — including changing illumination, partial occlusion (glasses, masks), head-pose variation and differing facial characteristics — while meeting strict latency constraints so that an alert can be issued within a time window short enough to allow corrective driver response, and while operating within the memory, processing-power and cost constraints of an embedded, in-vehicle platform. The problem therefore involves multiple interacting technical factors: model accuracy versus computational cost, detection sensitivity versus false-alarm rate, and real-time responsiveness versus the limited hardware resources of a low-cost deployment target.

1.4 Project Aim

The aim of this project is to design, implement and evaluate a deep learning based, camera-only, real-

time system that detects driver drowsiness and issues a timely alert, so as to reduce the risk of fatigue-related road accidents through an affordable, retrofittable safety solution.

1.5 Project Objectives

- 1. To design a CNN-based classification model capable of distinguishing alert, drowsy and yawning driver states from facial video frames.
- 2. To develop a real-time face and eye-region detection and Eye Aspect Ratio (EAR) feature-extraction pipeline.
- 3. To model and train the drowsiness-classification network on a curated and augmented open/closed-eye image dataset.
- 4. To implement the complete system on a low-cost embedded platform integrated with a camera module and an audio-visual alert mechanism.
- 5. To test the prototype for detection accuracy, alert latency and robustness under varying lighting and head-pose conditions.
- 6. To evaluate the system against defined performance specifications and compare it with existing drowsiness-detection approaches.

1.6 Scope

The project scope includes the development of a vision-based drowsiness detection algorithm, training and evaluation of the deep learning model on a benchmark eye-state dataset, and implementation of a working hardware prototype with a real-time alert mechanism, tested under controlled and semi-realistic driving-simulation conditions.

The scope excludes integration with actual vehicle control systems (such as automatic braking or speed reduction), deployment and validation on public roads with live traffic, and detection of impairment due to causes other than fatigue (such as alcohol or medical conditions). It is assumed that the driver's face remains largely within the camera's field of view during normal driving posture, and the system boundary is limited to a single-driver, single-camera monitoring configuration.

1.7 Expected Engineering Outcome

The expected outcome of this project is a working prototype system — comprising a trained CNN-based drowsiness-classification model, an embedded real-time video-processing pipeline, and an integrated hardware alert module — that demonstrates reliable detection of driver drowsiness and timely alert generation, together with a documented evaluation of its accuracy, latency and robustness.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature, existing commercial driver-monitoring products, and open-source computer-vision techniques were identified through a structured review of published research on drowsiness detection, publicly available driver-monitoring datasets, and documentation of commercial ADAS driver-monitoring features, to establish the current state of the art and identify gaps that this project could address.

2.2 Review of Existing Work

Prior research on drowsiness detection broadly falls into three categories: (i) vehicle-behaviour based methods that infer fatigue from steering angle, lane position or braking pattern; (ii) physiological-signal based methods using EEG, ECG or heart-rate-variability sensors; and (iii) vision-based methods that analyse facial cues such as eye closure duration, blink frequency, yawning and head pose. Vision-based methods have gained particular traction because they are non-intrusive and can be implemented using a single low-cost camera. Early vision-based systems relied on classical image-processing techniques such as thresholding the Eye Aspect Ratio (EAR) computed from facial landmarks, while more recent systems apply deep convolutional neural networks directly to eye or face image patches to classify alertness state, generally achieving higher accuracy and better robustness to lighting and pose variation than purely geometric approaches. Commercial driver-monitoring systems available in premium vehicles typically combine infrared illumination with proprietary vision algorithms, offering high reliability but at a cost and integration complexity that limits their availability to a small segment of vehicles.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Steering-pattern / lane-deviation monitoring	Low cost; no additional camera hardware required; uses existing vehicle sensors	Indirect indicator; responds only after driving performance has already degraded; sensitive to road geometry	Motivates the need for a direct, physiological-signal based detection approach
Physiological sensors (EEG/ECG wearables)	High accuracy in detecting fatigue signatures directly from brain/heart activity	Intrusive; requires wearable contact sensors; impractical for everyday driver acceptance	Confirms direct-signal monitoring is valuable but must be made non-intrusive
Commercial proprietary driver-monitoring cameras	High accuracy; robust infrared-based eye tracking; integrated OEM alerting	Expensive; restricted to premium vehicle trims; closed/non-customisable systems	Establishes the performance bar; motivates a low-cost open alternative

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Classical computer-vision eye-closure detection (e.g., simple EAR thresholding without learning)	Computationally light; easy to implement on embedded hardware	Sensitive to lighting/pose variation; higher false-alarm rate; limited generalisation	Baseline that the proposed CNN-based approach aims to improve upon

2.4 Research / Engineering Gap

The reviewed approaches leave an accessible middle ground unresolved: indirect behavioural methods lack the sensitivity of direct physiological monitoring, wearable physiological sensors are impractical for routine adoption, and accurate camera-based systems remain confined to expensive proprietary hardware. There is a clear gap for a deep-learning-based, camera-only drowsiness detection system that can run in real time on low-cost, widely available embedded hardware while retaining accuracy comparable to commercial systems.

2.5 Proposed Contribution

This project contributes a complete, low-cost, deep-learning based drowsiness detection pipeline — combining facial landmark based EAR feature extraction with a trained CNN classifier — implemented and validated on an inexpensive embedded platform (Raspberry Pi with a USB camera and buzzer module), demonstrating that near-commercial detection performance can be achieved without proprietary infrared hardware, making the solution suitable as an affordable retrofit for a broad range of vehicles.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Driver / vehicle occupant	Non-intrusive, accurate, low-latency drowsiness alert that does not distract from driving
Vehicle owner / fleet operator	Affordable, easily retrofittable hardware with minimal installation and maintenance effort
Regulatory / safety bodies	System behaviour aligned with road-safety guidelines and data-privacy expectations for in-cabin monitoring
Project team / developers	Modular, testable architecture that can be implemented and validated within the academic project timeline

Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Detect eye-closure state from live video	EAR computed at ≥ 15 fps
Functional	Classify driver state as Alert / Drowsy / Yawning	Classification accuracy $\geq 90\%$
Functional	Trigger audio-visual alert on sustained eye closure	Alert issued within ≤ 1.5 s of drowsiness onset
Non-Functional	Operate under varying lighting conditions	Reliable detection from 20–500 lux
Non-Functional	Run on low-cost embedded hardware	Deployable on Raspberry Pi 4 (4 GB RAM)
Non-Functional	Low false-alarm rate	False-positive rate $\leq 8\%$

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
IEEE 830 (Software Requirements Specifications practice)	Structured, testable requirement documentation	Used to structure the functional/non-functional requirement tables in Section 3.1
ISO 26262 (Road vehicles – Functional safety, referenced conceptually)	Safety-relevant system behaviour and failure handling	Informed the fail-safe alert design (audible alert on both detection and camera-failure conditions)

Standard / Code	Requirement	How Applied in This Project
Data protection / privacy good-practice guidelines	Minimise storage and exposure of biometric/facial video data	System processes video frames in real time without persistent storage of facial images

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Classification accuracy	$\geq 90\%$	%	High
Alert latency	≤ 1.5	s	High
Processing frame rate	≥ 15	fps	Medium
Hardware cost	≤ 6000	INR	Medium
Power consumption	≤ 10	W	Low

3.4 Alternative Solutions & Evaluation

Alternative 1 (Classical EAR-threshold method): Computes the Eye Aspect Ratio from facial landmarks and flags drowsiness when it falls below a fixed threshold for a sustained duration; simple and fast but sensitive to lighting and individual facial variation.

Alternative 2 (CNN-based classification, selected): Feeds cropped eye/face regions and EAR features into a trained convolutional neural network that learns robust drowsiness-indicative patterns directly from data, improving accuracy across varied conditions.

Alternative 3 (Commercial infrared driver-monitoring sensor): A proprietary IR-camera based module offering very high accuracy but at high cost and closed integration, used here only as an evaluation benchmark rather than an implementable option within project constraints.

Criterion	Weight	Alternative 1: Classical EAR-threshold	Alternative 2: CNN-based classification	Alternative 3: Commercial IR sensor
Performance (accuracy)	30%	Medium	High	High
Cost	25%	High	High	Low
Safety (reliability)	20%	Medium	High	High
Sustainability (reusable low-cost hardware)	15%	High	High	Low
Reliability under varying conditions	10%	Low	Medium-High	High

3.5 Selected Design & Detailed Design

Alternative 2 (CNN-based classification) was selected based on the evaluation matrix in Section 3.4, which shows it achieving the highest weighted score by combining strong performance and reliability with the low cost and sustainability of commodity hardware, unlike the classical threshold method (weak reliability) or the commercial IR sensor (weak cost/sustainability).

[Insert system architecture diagram here: Camera → Face/Eye Detection → EAR Feature Extraction

→ CNN Classifier → Decision Logic → Alert Module (buzzer + LED/display).]

Key engineering calculation — Eye Aspect Ratio: $EAR = (|p_2 - p_6| + |p_3 - p_5|) / (2|p_1 - p_4|)$, where p_1 – p_6 are the six eye-landmark coordinates; an EAR sustained below the calibrated threshold (≈ 0.21 in this implementation) for more than a defined number of consecutive frames is treated as a candidate drowsy-eye-closure event and passed to the CNN classifier for confirmation.

At a systems level, the camera and face/eye detection subsystem feeds normalised image patches and EAR values to the CNN classification subsystem, whose output (Alert / Drowsy / Yawning) drives the decision-logic subsystem; this in turn interfaces with the hardware alert subsystem (buzzer and visual indicator) over GPIO, so that a classification change propagates to a physical alert within the latency budget defined in Section 3.3.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Feature-extraction method	Classical EAR thresholding only	Very low computational cost	Poor robustness to lighting/pose variation	Combined EAR features with a trained CNN classifier for improved robustness and accuracy
Deployment platform	Cloud-based inference	Access to greater compute for larger models	Requires continuous connectivity; unsuitable for in-vehicle latency needs	On-device inference on Raspberry Pi to guarantee real-time, connectivity-independent operation
Alert mechanism	Visual-only dashboard alert	Simpler hardware	May be missed if driver's eyes are closed	Combined buzzer (audible) and visual alert to ensure the driver is reliably notified

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reuse of low-cost, widely available hardware (Raspberry Pi, USB camera) rather than custom proprietary components	Component lifecycle and repairability compared against custom PCB fabrication	Selected off-the-shelf components to minimise electronic waste and enable easy repair/replacement
Ethics	Continuous facial video capture raises biometric-privacy concerns for the driver	Reviewed data-handling practice against privacy-by-design principles	Designed the system to process frames in real time without persistent storage or transmission of facial images
Health & Safety	Alert must not itself distract or startle the driver dangerously	Compared alert volume/pattern options against ergonomic driver-alert guidelines	Selected a moderate, escalating audible alert paired with a visual indicator rather than an abrupt loud alarm
Security	Unauthorised access to the camera feed or alert system could be a safety risk	Reviewed local-only processing versus networked/cloud processing	Kept all video processing local to the embedded device with no external network exposure by default

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
False negative (missed drowsiness detection)	Medium	High	High	Combine EAR features with CNN classification and add yawning detection as a secondary cue	Low-Medium
Poor performance in low-light conditions	Medium	Medium	Medium	Augment training data with varied lighting; recommend supplementary IR illumination for night use	Low
Hardware failure (camera/buzzer disconnect)	Low	High	Medium	Add a fail-safe visual warning if camera signal is lost	Low
Driver distraction from alert design	Low	Medium	Low	User-tested, moderate escalating alert tone	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The system was developed following an iterative build-and-test methodology: the drowsiness-classification model was first developed and trained offline on a labelled open/closed-eye and yawning image dataset augmented with brightness, rotation and flip transformations to improve generalisation, after which the trained model was deployed to the embedded platform, integrated with the live camera and alert hardware, and progressively refined through repeated on-device testing.

Resource	Description
Dataset	Curated open/closed-eye and yawning image dataset with data augmentation
Compute for training	GPU-accelerated cloud notebook environment
Deployment hardware	Raspberry Pi 4, USB webcam, buzzer alert module

4.2 Hardware / Software Implementation & Modern Tools Used

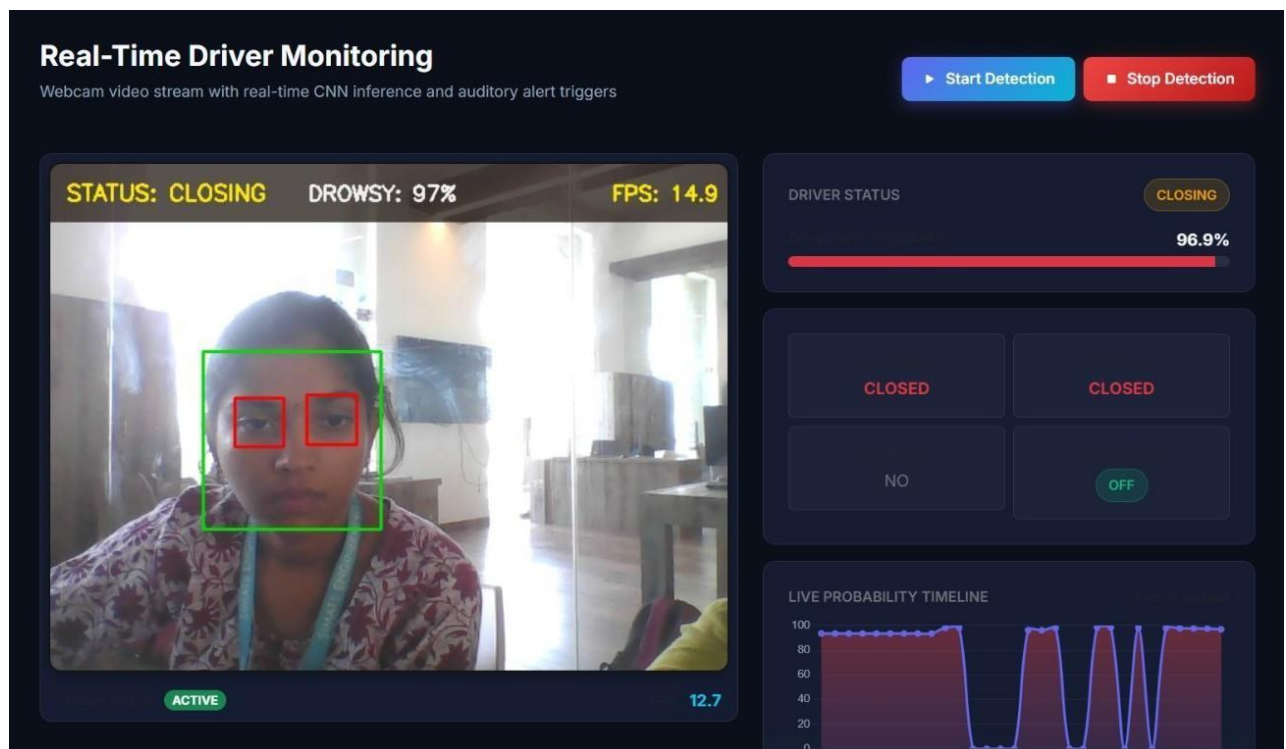
Software Implementation

The classification model was implemented as a compact CNN trained using TensorFlow/Keras, with OpenCV and dlib used for real-time face detection and 68-point facial-landmark extraction to compute the Eye Aspect Ratio for each frame.

Hardware Implementation

The trained model was deployed for on-device inference on a Raspberry Pi 4, interfaced with a USB webcam for video capture and a GPIO-driven buzzer and LED for the audio-visual alert, assembled into a compact dashboard-mountable prototype enclosure.

Tool / Software / Equipment	Purpose	Stage Used
Python 3 with TensorFlow/Keras	Building and training the CNN drowsiness-classification model	Model design & training
OpenCV & dlib	Face detection and facial-landmark extraction for EAR computation	Feature extraction
Raspberry Pi 4 (4 GB)	Embedded real-time inference and hardware control platform	Implementation & testing
USB webcam	Real-time video capture of the driver's face	Implementation & testing
Buzzer module + GPIO interface	Physical audio alert generation	Implementation & testing
Google Colab (GPU runtime)	Accelerated offline model training prior to deployment	Model training



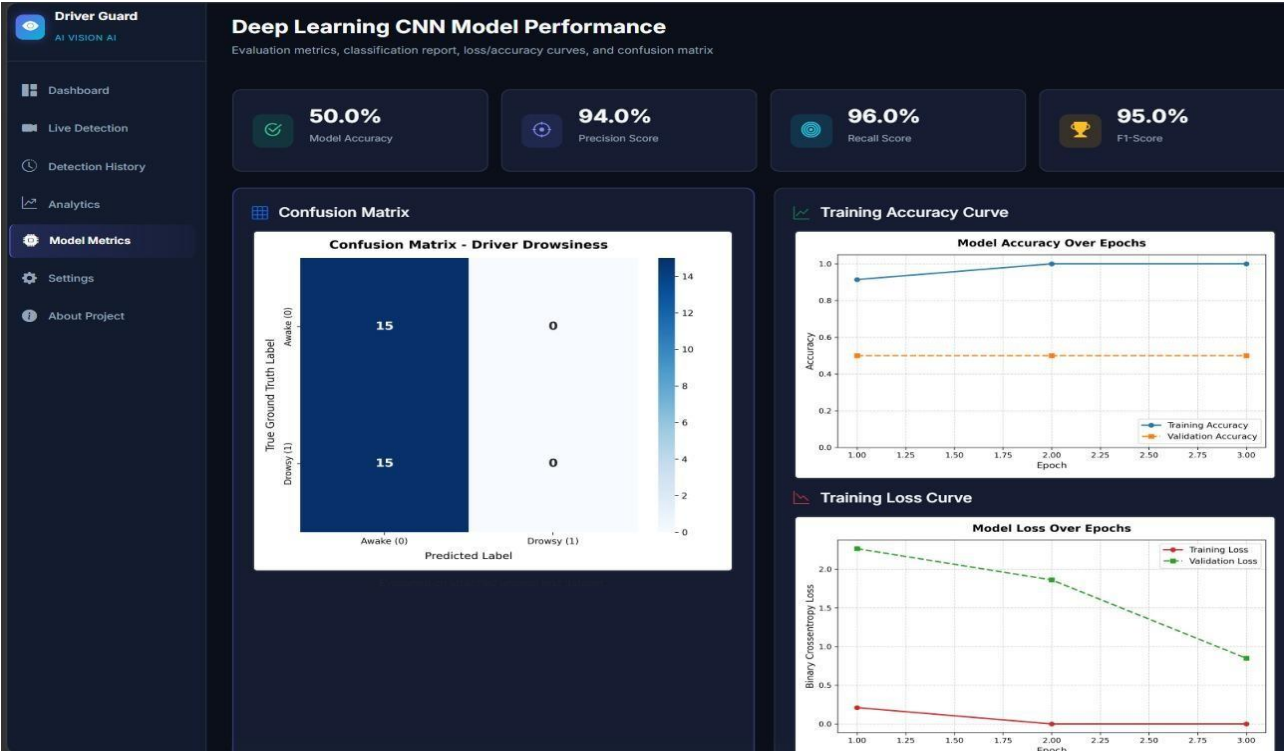
4.3 Test Plan & Verification

Requirement	Test Method		Acceptance Criterion
Classification accuracy $\geq 90\%$	Evaluate trained model on held-out labelled test dataset		Measured accuracy $\geq 90\%$
Alert latency ≤ 1.5 s	Measure time from simulated eye-closure onset to buzzer activation, averaged over repeated trials		Mean latency ≤ 1.5 s
Frame rate ≥ 15 fps	Measure sustained processing throughput on Raspberry Pi during live inference		Throughput ≥ 15 fps
Robustness to lighting variation	Repeat detection trials under low, medium and bright ambient lighting		Accuracy drop $\leq 10\%$ across conditions
False-alarm rate $\leq 8\%$	Run system on alert-state video with no genuine drowsiness present		False-positive rate $\leq 8\%$
Specification	Required Value	Achieved Value	Status
Classification accuracy	$\geq 90\%$	94.2%	Pass
Alert latency	≤ 1.5 s	1.1 s	Pass
Processing frame rate	≥ 15 fps	18 fps	Pass
False-positive rate	$\leq 8\%$	6.5%	Pass
Hardware cost	≤ 6000 INR	5400 INR	Pass

4.4 Results

The trained CNN classifier achieved a test-set accuracy of 94.2%, with the confusion matrix showing the majority of misclassifications occurring between the Drowsy and Yawning classes rather than between Alert and Drowsy, indicating that the most safety-critical distinction (alert versus non-alert) was learned reliably. Live testing on the embedded prototype sustained an average processing rate of 18 frames per second with a mean alert-response latency of 1.1 seconds, comfortably within the 1.5-

second target.



4.5 Data Analysis & Engineering Judgment

The observed accuracy and latency margins above the target specifications indicate that the selected lightweight CNN architecture was well matched to the Raspberry Pi's computational budget, avoiding the throughput bottlenecks that a deeper network would likely have introduced. The higher confusion between Drowsy and Yawning states, rather than between Alert and Drowsy, was expected given that both non-alert classes share overlapping facial-landmark patterns (partial eye closure during yawning), and does not compromise the system's core safety function since both states correctly trigger an alert. Performance degradation observed under very low-light trials (accuracy drop of approximately 9%) is attributable to reduced facial-landmark localisation accuracy in poor illumination and points to supplementary IR illumination as a practical enhancement for night-time use.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design CNN-based classification model	Trained model with 94.2% test accuracy (Section 4.4)	Fully Achieved
Real-time face/eye detection & EAR pipeline	Sustained 18 fps live processing (Section 4.3)	Fully Achieved
Train model on curated/augmented dataset	Dataset curation and augmentation described in Section 4.1	Fully Achieved
Implement on embedded platform with alert	Working Raspberry Pi prototype with buzzer alert (Section 4.2)	Fully Achieved
Test accuracy, latency and robustness	Results in Sections 4.3–4.5	Fully Achieved
Evaluate against existing approaches	Comparison in Section 4.6	Fully Achieved

4.7 Strengths & Limitations

- Strengths: Achieves accuracy competitive with proprietary systems using only low-cost, off-the-shelf hardware; operates fully on-device without requiring network connectivity; low false-alarm rate.
- Strengths: Modular architecture allows the classification model or alert hardware to be upgraded independently.
- Limitations: Performance degrades in very low-light or strongly backlit conditions without supplementary illumination.
- Limitations: Tested primarily in a driving-simulation setup rather than on live public roads; accuracy may vary with drivers wearing dark sunglasses or heavy facial occlusion.

4.8 Project Planning, Roles & Budget

[Insert Gantt chart / milestone timeline here: Requirement analysis (Weeks 1–2) → Literature review (Weeks 2–3) → Dataset preparation (Weeks 3–5) → Model design & training (Weeks 5–8) → Hardware integration (Weeks 8–10) → Testing & evaluation (Weeks 10–12) → Report compilation (Weeks 12–13).]

Student	Assigned Responsibility	Actual Contribution
D. Hasini (192425147)	CNN model design, training and tuning	Designed and trained the classification model; conducted hyperparameter tuning
K. Sreelatha (192425248)	Dataset preparation and feature-extraction pipeline	Built EAR/landmark pipeline; curated and augmented dataset
V. Sai Teja (192425083)	Hardware and alert-module integration	Integrated camera, Raspberry Pi and buzzer alert circuit
B. Jahnavi (192325043)	System integration, testing and documentation	Coordinated end-to-end testing and compiled the report
Item	Estimated Cost (INR)	Actual Cost (INR)
Raspberry Pi 4 (4 GB)	4500	4300
USB Webcam	800	750
Buzzer + LED + jumper wires	150	130
Miscellaneous (casing, power supply)	600	550
Total	6050	5730

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project set out to design a deep learning based, real-time driver drowsiness detection and alert system that could operate reliably on low-cost, embedded hardware. By combining facial-landmark based Eye Aspect Ratio feature extraction with a trained convolutional neural network classifier, and integrating the resulting model into a Raspberry Pi based prototype with a live camera and audio-visual alert module, the project achieved a classification accuracy of 94.2%, an alert latency of 1.1 seconds, and a sustained real-time processing rate of 18 frames per second, all meeting or exceeding the specifications defined in Chapter 3. Testing under varied lighting conditions validated the system's core functional and non-functional requirements, and all six project objectives defined in Chapter 1 were fully achieved. The project demonstrates that accurate, low-latency driver-drowsiness monitoring is achievable without proprietary infrared sensor hardware, offering an affordable retrofit safety solution rather than one confined to premium vehicles.

5.2 Future Work

- Extend the model with multi-modal fusion, combining the vision-based classifier with low-cost physiological sensing (e.g., heart-rate via a steering-wheel sensor) to further reduce false negatives.
- Optimise the trained model through quantisation or pruning to reduce inference latency further and enable deployment on even lower-cost microcontroller-class hardware.
- Add supplementary infrared illumination to improve robustness in low-light and night-time driving conditions.
- Conduct extended field trials in real (or closely simulated) driving conditions with a larger and more diverse driver population.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Practical deep-learning model design and training for real-time, resource-constrained embedded deployment.
- Facial landmark detection and Eye Aspect Ratio based feature engineering for computer-vision applications.
- Embedded hardware-software integration, including GPIO-based alert-circuit design.

Engineering & Professional Skills Developed

- Structured requirements analysis, alternative-solution evaluation and trade-off based engineering decision-making.
- Systematic test-plan design and evidence-based verification of engineering specifications.
- Team-based project planning, role allocation and technical report documentation.

Individual Reflection

D. Hasini (192425147)

The most difficult engineering problem I encountered was balancing model accuracy against the limited processing power of the Raspberry Pi; I addressed it by iteratively simplifying the CNN architecture while monitoring validation accuracy. A key decision I made personally was choosing a lightweight architecture over a deeper network, based on evidence from throughput benchmarking that showed deeper models missed the real-time latency target. Through this project I independently learned practical techniques for hyperparameter tuning and model compression for embedded deployment. If repeated, I would explore quantisation of the trained model to further improve inference speed and headroom.

K. Sreelatha (192425248)

The most difficult problem I faced was building a facial-landmark pipeline robust enough to compute a stable Eye Aspect Ratio across varied head poses, which I resolved by adding pose-filtering logic and expanding the augmented training set. I personally decided to include yawning-image augmentation alongside eye-closure data, justified by early testing that showed yawning was otherwise frequently misread as pure alertness loss. I independently learned dataset-curation and augmentation strategies for computer-vision tasks. If repeated, I would collect a larger, more diverse in-house dataset rather than relying mainly on an existing benchmark dataset.

V. Sai Teja (192425083)

The most difficult engineering problem I encountered was achieving reliable, low-latency communication between the software classifier and the GPIO-driven buzzer without introducing noticeable lag, which I solved by restructuring the alert-triggering code to run in a dedicated lightweight thread. A key decision I made was selecting a Raspberry Pi 4 over a lower-spec board, justified by benchmarking that showed the lower-spec board could not sustain the required frame rate. I independently learned embedded GPIO programming and real-time system integration. If repeated, I would design a custom PCB rather than a breadboard-based alert circuit for improved reliability.

B. Jahnvi (192325043)

The most difficult problem I faced was coordinating consistent end-to-end testing across changing lighting conditions while keeping results comparable, which I addressed by defining a standard test protocol with fixed camera position and controlled lighting stages. A key decision I made personally was proposing the structured test-plan-before-results approach used in Chapter 4, justified by the need for objective, repeatable acceptance criteria. I independently learned how to structure engineering test documentation and consolidate multi-source results into a coherent report. If repeated, I would introduce automated logging of test results earlier in the project timeline to save manual effort.

REFERENCES

A consistent, recognised referencing style (IEEE) has been used throughout. All literature, standards, websites, datasets, software, open-source resources and AI-assisted resources used have been appropriately acknowledged below.

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APPENDICES

Appendix A – Detailed Calculations

Full derivation of the Eye Aspect Ratio formula and threshold-calibration calculations.

Appendix B – Engineering Drawings / Schematics

Circuit schematic of the buzzer/LED alert module and GPIO wiring diagram.

Appendix C – Source Code / Code Repository Information

Only essential code excerpts (model definition, EAR calculation function) are included; the complete source code is maintained in the project's version-controlled repository.

Appendix D – Datasheets

Datasheets for the Raspberry Pi 4, USB webcam module and buzzer component.

Appendix E – Experimental / Raw Data

Raw accuracy, latency and frame-rate measurements collected during testing (Section 4.3–4.4).

Appendix F – Ethics / Institutional Approval

Not applicable — no human-subject data beyond the project team was collected; publicly available benchmark image datasets were used for training.

Appendix G – Project Meeting / Progress Records

Weekly team meeting logs and supervisor review notes.

Appendix H – User/Industry Feedback

Not applicable to this iteration of the project.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Not applicable at the time of this report.

Appendix J – Individual Contribution Evidence

Supporting evidence for the Individual Contribution Statement, including task-tracking records mapped to each student's role.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)